ANGEL OZA

Software Engineering | Robotics & Automation

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SUMMARY

ICT undergraduate with experience in embedded systems and hardware prototyping, with hands-on project experience in microcontroller-based control, sensor integration, LED/display systems, and wireless communication and building practical projects. Have taken up leadership roles in student organizations, managing teams and organizing technical events. Comfortable with problem-solving, working in teams, and learning new tools as required

EDUCATION

MARWADI UNIVERSITY — **Aug 2023 - May 2027**

Bachelor of Technology - Information and Communication Technology

- Coursework: Data Structures and Algorithms, Object-oriented Programming, Database Management Systems, Operating Systems, Computer Networks, and Software engineering

Genius School — **April 2021 - May 2023**

Higher Secondary

- Percentage: 62%

PROJECTS

1. **Hand Gesture-Controlled Wheelchair**
   - Designed and developed an assistive mobility device for paralysis patients using gesture-based control with flex sensors.
   - Programmed a microcontroller for real-time motor control, ensuring reliable signal processing and responsiveness.
   - Focused on user-centric design by improving accuracy, comfort, and ease of use.
2. **Night Sky Display**
   - Developed an interactive night sky model using an Arduino-controlled LED matrix.
   - Implemented GPS and orientation-based tracking to render real-time constellation positions.
   - Applied graph-based mapping techniques for accurate star visualization.
   - Designed dynamic display patterns to illustrate mathematical concepts on the LED matrix.
3. **Sign Speak — Speech-to-Sign Language & Gesture Recognition System**
   - Developed a real-time assistive communication system that converts speech into sign language gestures and recognises hand gestures into text/speech using Python.
   - Implemented speech recognition, webcam-based gesture detection, and multilingual support using OpenCV, SpeechRecognition, NumPy, and machine learning techniques.
   - Designed an interactive browser-based interface for real-time communication support for deaf and mute individuals.
4. **SurakshaLink**
   - Developed an offline emergency communication system using ESP32 and LoRa.
   - Implemented structured wireless communication with alert prioritization.
   - Integrated acknowledgment handling for reliable node-to-node messaging.
   - Designed a real-time dashboard for communication monitoring and logs.

RESEARCH EXPERIENCE

- Developing and simulating THz-band MIMO antenna configurations in HFSS for spectroscopy and imaging applications.
- Analyzing S-parameters, return loss, and radiation characteristics to optimize antenna performance.
- Exploring data-driven methods for antenna parameter optimization and performance prediction.

SKILLS

Programming & Software Development

- Languages: C++, Python, Java
- Backend & APIs: REST APIs, Java (JSP/Servlets), MySQL
- Web Technologies: ReactJS, HTML, CSS, JavaScript

Systems & Performance

- Distributed & Multicore Systems, Performance Optimization, System Profiling
- Secure SDLC, DevSecOps, Agile Development Practices

Machine Learning (Supporting)

- Frameworks: PyTorch, TensorFlow
- Libraries: Pandas, NumPy, Matplotlib

Hardware & Embedded (Supporting)

- Computer Architecture, Digital Systems
- Embedded Systems: Arduino, Raspberry Pi, ESP8266/ESP32, Zigbee
- Circuit Design & Simulation: PCB Design, Logic Design

LEADERSHIP & STEM ACTIVITIES

- Chair - IEEE MEFGI Student Branch (2026): Led student operations, coordinated technical events, and collaborated with industry professionals.
- Convenor - Circuitology Club (2024 - '25): Built a 110+ member technical community, secured USD 3,500+ in funding, and organized 30+ technical sessions and STEM outreach programs.
- Marketing Lead - Google Developer Group On-Campus MU: As Marketing Lead at GDG on Campus, strategized and executed the chapters branding and communication initiatives, leading design, content, and outreach efforts to build a strong tech-driven community on campus.